

Towards Asynchronous BFT Micro Replication

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Abstract: The concept of micro replication offers significant benefits in terms of understandability, debuggability and resilience, but so far has only been explored in the context of partially synchronous systems. In this paper, we report on our ongoing efforts to leverage the approach also for asynchronous state-machine replication. Specifically we present MICRONEY, the first asynchronous Byzantine fault-tolerant (BFT) protocol that incorporates the micro-replication principles.

1 Introduction

Traditionally, system architectures for state-machine replication place the responsibilities for essential tasks (such as client interaction, the consensus process, or request processing) in the hands of monolithic replicas that each need to provide means to handle all of them [Di21]. The result of this design strategy is a highly complex protocol logic that is inherently difficult to understand and debug, and which entails the risk of containing vulnerabilities that are caused by different sub protocols interfering with each other in unexpected and unwanted ways.

Micro replication [DEL23] addresses these issues by requiring each step of a state-machine replication protocol to be carried out by a dedicated cluster of *micro replicas* which solely comprise the logic for one particular task. Studies show that this approach not only makes it easier to identify and avoid bugs [DEL23], it also can be used to efficiently increase the resilience of a replicated system by applying selective hybridization and diversification [LD24, LD25]; that is, to specifically protect the most vulnerable or most exposed protocol steps against Byzantine faults, and to decrease the probability of common-mode failures using heterogeneous implementations for the corresponding micro replicas.

With all previous works in this area addressing partially synchronous systems [DEL23, LD24, LD25], up to this point it remained an open question whether asynchronous systems may benefit from micro replication as well. To answer this question, we are currently developing MICRONEY, the first asynchronous Byzantine fault-tolerant micro-replication protocol.

2 Challenges

Designing a micro-replicated system architecture essentially requires to split a replication protocol into its atoms (i.e., protocol steps) and model each of them in its own micro replica. For some building blocks of asynchronous protocols, for example the mechanisms for communicating with clients, it is possible to exploit similarities to partially synchronous protocols and hence draw on already existing micro-replica specifications. In contrast, other parts are particularly challenging. Unlike most of their partially synchronous counterparts, asynchronous protocols for example rely on Byzantine agreement algorithms that do not reach consensus in a deterministic number of communication phases, which makes it not straightforward to identify the total number of micro replicas that the system needs to comprise.

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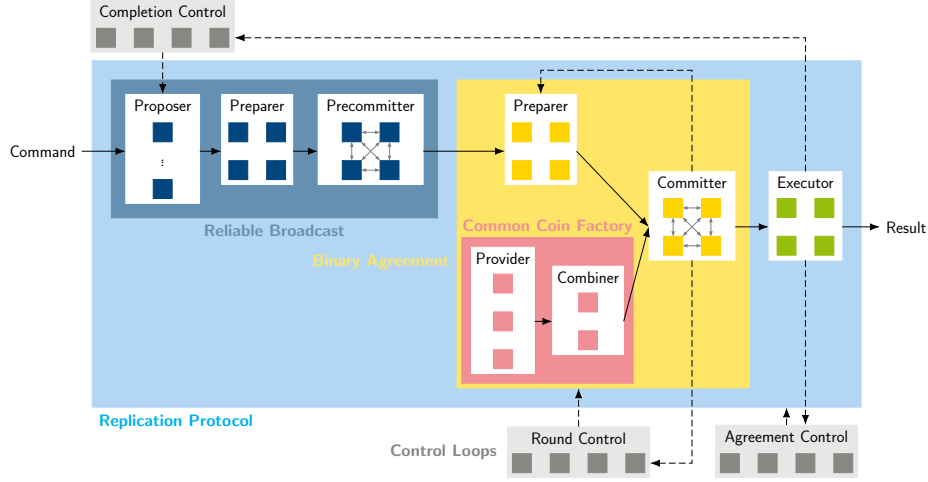


Fig. 1: MICRONEY overview

3 MICRONEY

MICRONEY’s system architecture is inspired by the workflow of HoneyBadgerBFT [Mi16] and, as illustrated in Figure 1, comprises three main stages. First, a set of micro-replica clusters implements a reliable broadcast to disseminate batches of incoming client commands. Next, a second stage relies on binary agreement to establish a total order on these command batches. Finally, a cluster of executor micro replicas processes the commands in the determined order and returns the corresponding results to clients. In addition to these main stages, MICRONEY contains three control loops which feed progress updates from downstream clusters back to upstream clusters and thereby enable garbage collection of no longer needed state.

In sum, MICRONEY is a composition of 11 micro-replica clusters, which each are responsible for a different step of the overall protocol. This specialization also makes it possible to tailor the size of each cluster to the individual characteristics of its protocol step, as for example shown by the clusters that produce the common coin required for binary agreement.

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